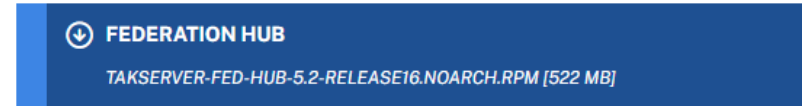


# TAK Server - Federation Hub

# TAK Server – Federation Hub Installation

- You should have your TAK Server already installed and running.
- Download Federation Hub from the following link:
  - <https://tak.gov/products/tak-server>
- Update package manager:
  - **sudo yum update -y**
- Ensure you are running Java version 17:
  - **java --version**
  - To install java 17:
    - **sudo dnf install java-17-openjdk-devel -y**



```
[takadmin@localhost ~]$ java --version
openjdk 17.0.12 2024-07-16 LTS
OpenJDK Runtime Environment (Red_Hat-17.0.12.0.7-1) (build 17.0.12+7-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-17.0.12.0.7-1) (build 17.0.12+7-LTS, mixed mode, sharing)
[takadmin@localhost ~]$
```

# TAK Server – Federation Hub Installation

- Copy the Federation Hub file over to the server:
  - On your local machines terminal:
    - **scp takserver-fed-hub-\*.rpm <username>@<IP address>:~/**
      - Example: **scp takserver-fed-hub-\*.rpm takadmin@192.168.1.123:~/**
      - You can use the entire file name if you have multiple versions downloaded.
- Install the RPM:
  - **sudo dnf install takserver-fed-hub-\*.noarch.rpm -y**
- Add and Apply SELinux:
  - **sudo dnf install checkpolicy**
  - **cd /opt/tak/federation-hub && sudo ./apply-selinux.sh && sudo semodule -l | grep takserver**

# TAK Server – Federation Hub Installation

- Add database credentials (otherwise the default will be generated):
  - Open the federation hub broker file as the tak bash user:
    - **sudo su tak**
    - **vim /opt/tak/federation-hub/configs/federation-hub-broker.yml**
    - Scroll down to “dbPassword” and enter in a valid password:
- Optionally install Mongo: (cannot be done as the bash user “**exit**”)
  - **sudo cp /opt/tak/federation-hub/scripts/db/mongodb-org.repo /etc/yum.repos.d/mongodb-org.repo**
  - **sudo yum install -y mongodb-org**
  - **sudo systemctl daemon-reload**
  - **sudo systemctl enable mongod**
  - **sudo systemctl restart mongod**
  - **sudo /opt/tak/federation-hub/scripts/db/configure.sh**

```
id: "6ac1016eb7ae44428050f8eefb077565"  
dbUsername: "martiuser"  
dbPassword: "atakatak"  
dbPort: 27017  
dbHost: "localhost"
```

# TAK Server – Federation Hub Firewall Configuration

- Configure the necessary ports:
  - **sudo firewall-cmd --zone=public --add-port 9100/tcp --permanent**
  - **sudo firewall-cmd --zone=public --add-port 9102/tcp --permanent**
  - **sudo firewall-cmd --reload**
  - You can view the current port configurations by running the following:
    - **sudo firewall-cmd --list-all**

```
[takadmin@localhost ~]$ sudo firewall-cmd --list-all
[sudo] password for takadmin:
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s8
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8089/tcp 8443/tcp 8446/tcp 9100/tcp 9102/tcp
  protocols:
  forward: no
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[takadmin@localhost ~]$
```

# TAK Server – Federation Hub Configuration

- Create your certificates:
  - Switch to the tak bash user:
    - **sudo su tak**
  - Navigate to the cert directory:
    - **cd /opt/tak/federation-hub/certs**
  - Modify the meta data file:
    - **vim cert-metadata.sh**
    - Update the values for the following:
      - **COUNTRY:** XX
      - **STATE:** XX
      - **CITY:** XX
      - **ORGANIZATION:** XX
      - **ORGANIZATIONALUNIT:** XX
  - Save and exit: press **esc**
    - type: **:wq!**
    - press enter

```
# Common configuration for all certificates
# Edit these fields to be appropriate for your organization
# If they are left blank, they will not be included. Do not leave COUNTRY
# blank (you may set it to "XX" if you want to be obtuse).
#
# Values for each may be optionally set as environment variables.
# Replace variables such as ${STATE} and ${CITY} as needed.
#
COUNTRY=US
STATE=WW
CITY=WW
ORGANIZATION=WW
ORGANIZATIONAL_UNIT=WW

CAPASS=${CAPASS:-atakatak}
PASS=${PASS:-$CAPASS}

## subdirectory to put all the actual certs and keys in
DIR=files
```

# TAK Server – Federation Hub Configuration

- Create the certificates needed for federation hub:
  - Create the root certificate:
    - **`./makeRootCa.sh --ca-name <name>`**
      - Example: **`./makeRootCa.sh --ca-name truststore-root`**
  - Create the server certificate:
    - **`./makeCert.sh server <name>`**
      - Example: **`./makeCert.sh server takserver`**

# TAK Server – Federation Hub Configuration

- Verify and update the file paths for the certificates:
  - Switch to the tak bash user:
    - **sudo su tak**
  - Update the path for the root certificate file if not the default (**truststore-root**):
    - **sed -i 's/truststore-root/truststore-<CACommonName>/g' /opt/tak/CoreConfig.example.xml**
      - Example: **sed -i 's/truststore-root/truststore-ROOT-CA-01/g' /opt/tak/federation-hub/configs/federation-hub-ui.yml**
- Update both of the paths for the server certificate file if not the default (**takserver**):
  - **sed -i 's/takserver/<TAKServerName>/g' /opt/tak/federation-hub/configs/federation-hub-broker.yml**
  - **sed -i 's/takserver/<TAKServerName>/g' /opt/tak/federation-hub/configs/federation-hub-ui.yml**



# TAK Server - Federation Hub - Configuring Authorized Users

- By default, authorized users are written to **/opt/tak/federation-hub/authroized\_users.yml** but can be written to an alternative location by running the following:
  - **java -jar /opt/tak/federation-hub/jars/federation-hub-manager.jar <path to certs>/cert.pem <path>/authorized\_users.yml**
  - Note: If you change the location of where the file is saved you will also need to update this location within the admin UI microservice:
    - **/opt/tak/federation-hub/configs/federation-hub-ui.yml**

# TAK Server - Federation Hub - Configuring Authorized Users

- Federation hub authenticates clients using TLS (Transport Layer Security) with X509 client certificates.
- If you need to create a new certificate you can run the following first:
  - **sudo su tak**
  - **cd /opt/tak/federation-hub/certs**
  - **./makeCert.sh client admin**
- To authorize clients to act as administrators and enable access to the admin UI run the following:
  - **sudo su tak**
  - **java -jar /opt/tak/federation-hub/jars/federation-hub-manager.jar <path to certs>/cert.pem**
    - Example: **java -jar /opt/tak/federation-hub/jars/federation-hub-manager.jar /opt/tak/federation-hub/certs/files/admin.pem**
  - **exit**

# TAK Server – Federation Hub Startup

- Start and enable the system service:
  - **sudo systemctl daemon-reload**
  - **sudo systemctl restart federation-hub**
  - **sudo systemctl enable federation-hub**
- You can also stop the service by running the following:
  - **sudo systemctl stop federation-hub**

# TAK Server - Federation Hub - Configuring Authorized Users

- Downloading admin certificate:
  - Ensure you are not in the tak user anymore:
    - **exit**
  - Return to the home directory:
    - **cd ~**
  - Copy the user certificate to your home directory:
    - **sudo cp -v /opt/tak/federation-hub/certs/files/<name>.p12 .**
      - Example: **sudo cp -v /opt/tak/federation-hub/certs/files/admin.p12 .**
  - Change the ownership to our Linux user
    - **sudo chown -R <username>:<usergroup> /home/<username>**
      - Example: **sudo chown -R takadmin:takadmin /home/takadmin**
  - Download the certificate to our local machine:
    - From a new terminal on our local machine run the following:
      - **scp <username>@<ipaddress>:~/<user>.p12 .**
        - Example: **scp [takadmin@192.168.76.246:~/admin.p12](#) .**

# TAK Server - Federation Hub - Configuring Authorized Users

- Import the user certificate into your browser of choice:
  - Open chrome/edge and navigate to the browser settings.
  - Search for “Manage Certificates” and select the setting.
  - Select “Import” → Select “Next”.
  - Select Browse and navigate to the downloaded certificate.
  - Open the certificate → Select “Next”.
  - Enter the password: default if “atakatak”.
  - Select “Next” → Select “Next” → Select “Finish”

# TAK Server - Federation Hub - Keycloak

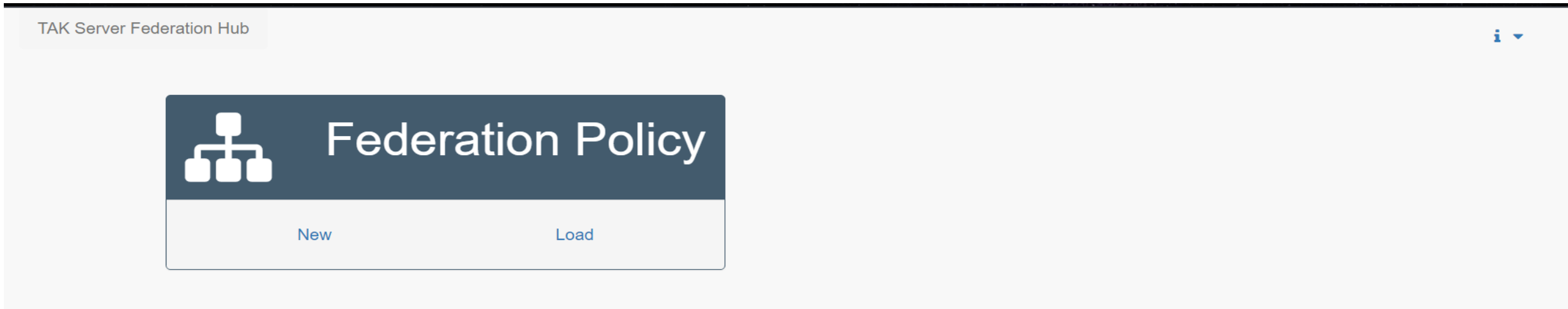
- Federation Hub supports OAuth admin authentication using keycloak, which can be configured by adding the following to the **federation-hub-ui.yml** file:
  - **vim /opt/tak/federation-hub/configs/federation-hub-ui.yml**

```
allowOAuth: true
oauthPort: 9099 #make sure this matches the redirect uri
keycloakServerName:
keycloakDerLocation: /opt/tak/federation-hub/certs/keycloak-certs.der
keycloakClientId:
keycloakSecret:
keycloakRedirectUri: https://localhost:9099/login/redirect #make sure this matches the oauth port
keycloakAuthEndpoint:
keycloakTokenEndpoint:
keycloakAccessTokenName:
keycloakRefreshTokenName:
keycloakClaimName: groups #the keycloak token attribute to look at for authenticating
keycloakAdminClaimValue: admin #the keycloakClaimName value to match for verifying admin status
```

---

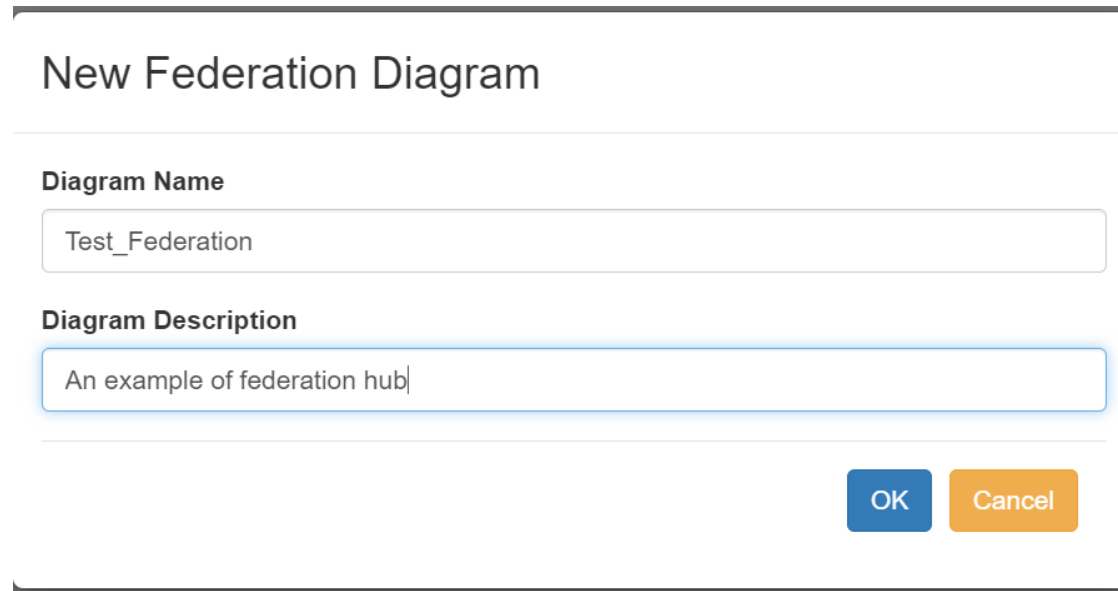
# TAK Server - Federation Hub - How to Use

- In a web browser navigate to `https://<server ip>:9100` (you can also use “localhost” if you are accessing this from the server).
  - Example: <https://192.168.76.246:9100>
  - To authenticate to the Federation Hub, you should be using the same certificate that was authorized in the “Configuring Authorized Users” section.
- When you successfully connect you will see the following page:



# TAK Server - Federation Hub - How to Use

- Select “New” and fill in the required information.
  - Select “OK” when done.



The screenshot shows a dialog box titled "New Federation Diagram". It contains two text input fields. The first field, labeled "Diagram Name", contains the text "Test\_Federation". The second field, labeled "Diagram Description", contains the text "An example of federation hub". At the bottom right of the dialog box are two buttons: "OK" (blue) and "Cancel" (orange).

New Federation Diagram

Diagram Name

Test\_Federation

Diagram Description

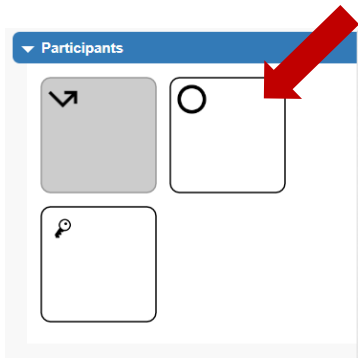
An example of federation hub

OK Cancel



# TAK Server - Federation Hub - How to Use

- Adding CA Groups:
  - Add a CA (Certificate Authority) to the empty policy, this will be the CA of a TAK Server that we want to connect to the Federation Hub.
    - To do this we will need the **ca.pem** file from the TAK Server:
      - From the server: **sudo cp -v /opt/tak/certs/files/ca.pem .**
        - **sudo chown -R takadmin:takadmin /home/takadmin**
      - From your local machine: **scp [takadmin@192.168.1.123:~/ca.pem](#) .**
    - You can also copy the file from you local machine by running the following:
      - **scp takadmin@192.168.1.123:/opt/tak/certs/files/ca.pem .**
  - In the Federation Hub editor drag and drop the group icon to the canvas:



# TAK Server - Federation Hub - How to Use

- In the form that popped up, fill in the following:
  - Select “Browse” and select the ca.pem file you copied over from the TAK Server.
  - Once selected press “Save”
  - Select the drop-down option for “CA Group (issuer)” and select the group corresponding to your CA. This will populate the other fields.
  - You will also want to enable the “Interconnected” check box option, which allow TAK Servers within the same group to communicate with each other.
  - You may also provide a graph name and short description.
  - Select “OK” when you are done to add the group policy.

Add Group

[Remove CA's](#)

Federate Identity

Graph Name

Description

CA Group (issuer)

CA Group (subject)

Interconnected ☒

[OK](#) [Cancel](#)

Add CA

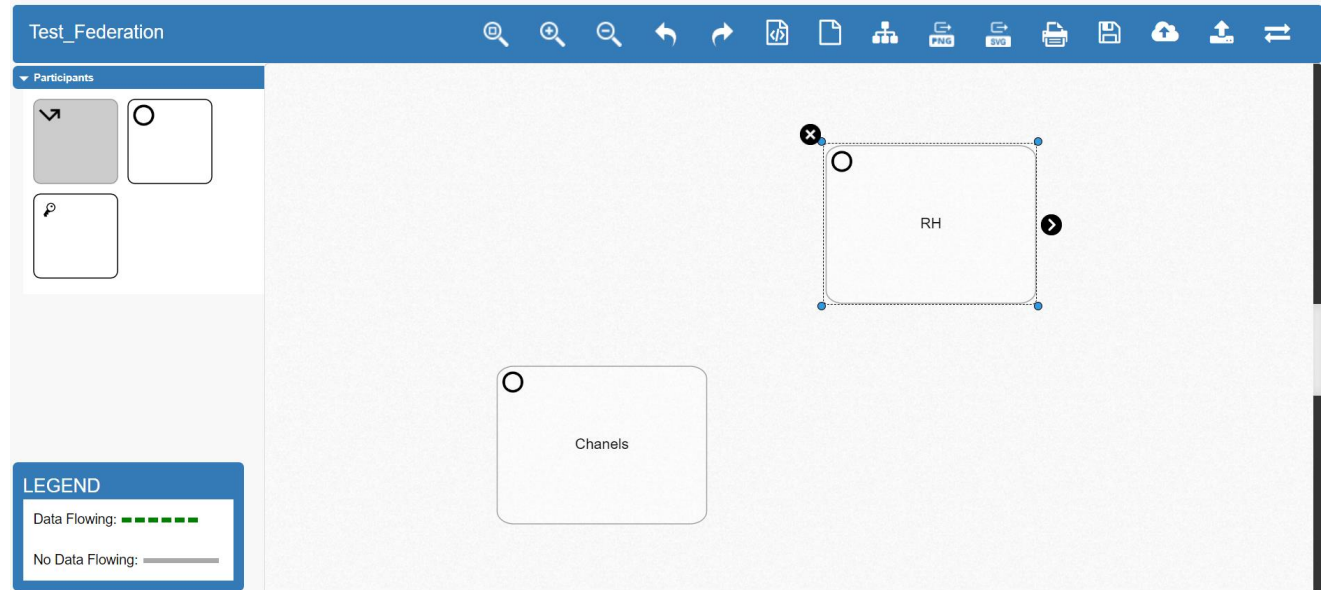
[Choose File](#) ca.pem

To add a CA, upload its X.509 certificate in .pem format.  
Warning: adding a CA will restart the federation server.

[Cancel](#) [Save](#)

# TAK Server - Federation Hub - How to Use

- Repeat the previous steps to add an additional CA.
  - If done correctly it should look like →
- Note: You can also add individual federates too the policy using the federates identity but it is not recommended.



# TAK Server - Federation Hub - How to Use

- To connect CA Groups, click on either of the groups once to select it, showing a small arrow icon on the right of the select group.
- Click and drag the icon to another group to connect them both, which will open a form “Add Federate Policy”
  - From here, group filtering can be configured for messages going from one node to another if the originator of the message has group mapping enabled on their TAK Server.
  - Filtering Options:
    - **Allow All Messages:** must be used for all V1 connections and any V2 connections that aren't running group mapping.
    - **Allowed Groups:** allow any message containing one of these groups.
    - **Disallowed Groups:** block any message containing one of these groups.
    - **Allowed and Disallowed Groups:** allow messages that contains an allowed group and contains no disallowed groups.
  - You can double click on the connection to change the group filtering options.

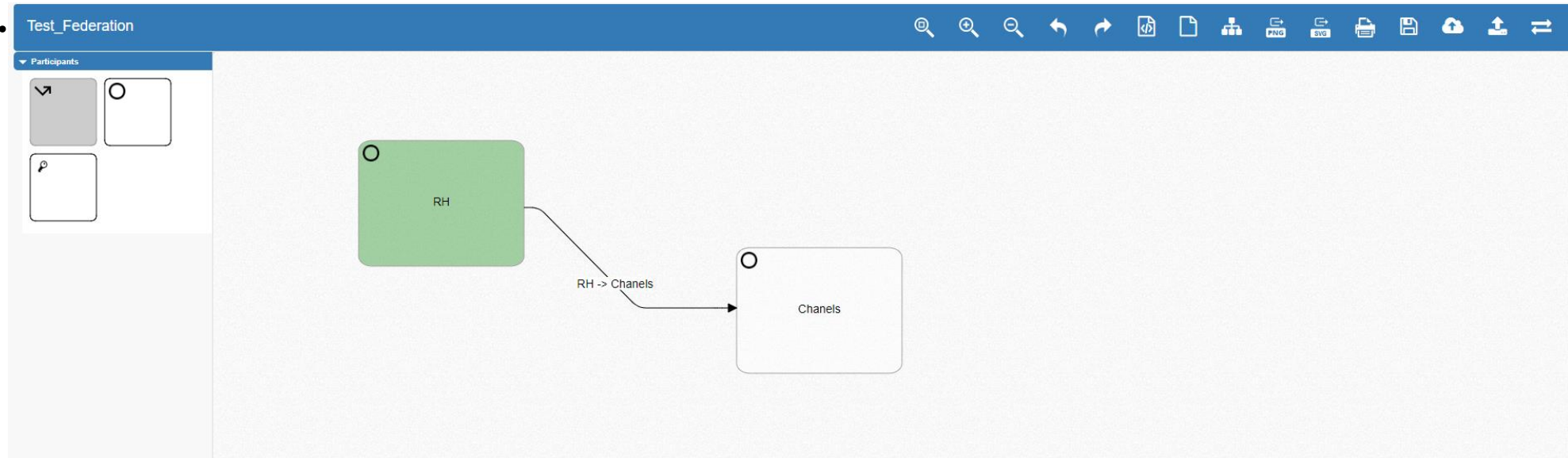
# TAK Server - Federation Hub - How to Use

- To upload the policy to the Federation Hub select the “Set as Active Policy and Save to Server” icon in the top right. (cloud icon)
  - This will make the configuration the active policy of the Hub and save it to the policy file that is used by the Hub located at: /opt/tak/federation-hub/ui\_generated\_policy.json
- If a certificate authority was added during a policy construction process you should restart the Federation Hub to load the latest truststore:
  - **sudo service federation-hub restart**



# TAK Server - Federation Hub - Connecting Federates

- Once the policy is configured, federates that are present in the policy can connect, while others will not be able to.
- To add a federation connection to the Federation Hub you will need to use the IP address of the Hub instead of another TAK server and use the Federation Hub's ports (9100 & 9102 for v2).
- When a successful connection to the Hub is made the group node will turn green.



# TAK Server - Federation Hub - Connection Walkthrough

- Download the **ca.pem** certificate for the federation hub.
  - Server: **sudo cp -v /opt/tak/federation-hub/certs/files/ca.pem .**
  - Local: **scp <username>@<ip address>:~/ca.pem .**
- Open your TAK Server admin dashboard and go to the following:  
Hamburger menu → Configuration → Federation Certificate Authority
- Select “Upload Federate Certificate Authority”.
- Select the file and save.

# TAK Server - Federation Hub - Connection Walkthrough

- Open your TAK Server admin dashboard and go to the following: Hamburger menu → Configuration → Federation
- From there select “Edit Configuration” and enable “Federation Enabled:”.
  - Select Save at the bottom.
- The page should update, and you should now see the following: →

## Active Connections

No Active Connections

## Outgoing Connection Configuration

[Create Outgoing Connection](#)

No Outgoing Connections

## Federate Configuration

No Federates Configured

## Federation Configuration

|                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Federation Enabled:</b> <input checked="" type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                       |
| <b>Federation V1 Enabled:</b> <input type="checkbox"/><br>Core Messaging Version: 2<br>Port: 9000, TLS Version(s): TLSv1.2, TLSv1.3                                                                                                                                                                                                                                                                                  |
| <b>Federation V2 Enabled:</b> <input checked="" type="checkbox"/><br>Port: 9001<br><b>Allow Mission Federation:</b> true<br><b>Allow DataFeed Federation:</b> true<br><b>Allow Federated Delete:</b> false<br><b>Federated Group Mapping:</b><br><b>Automatic Group Mapping:</b><br><b>Enable Mission Federation Disruption Tolerance:</b> true<br><b>Mission Federation Disruption Tolerance Interval:</b> 12 hours |
| <a href="#">Clear Federation Events</a>                                                                                                                                                                                                                                                                                                                                                                              |
| Will resync federated missions on reconnect                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Enable Data Package And Mission File Filter:</b> false                                                                                                                                                                                                                                                                                                                                                            |
| <b>Federation Truststore File Path:</b> certs/files/fed-truststore.jks                                                                                                                                                                                                                                                                                                                                               |
| <b>Web Base URL:</b> https://127.0.0.1:8443/Marti                                                                                                                                                                                                                                                                                                                                                                    |

[Edit Configuration](#)



# TAK Server - Federation Hub - Connection Walkthrough

- From the page select “Create Outgoing Connection” and fill in the following:
  - **Display Name:** <your choice>
  - **Address:** <IP of the Federation Hub>
  - **Port:** <9100 or 9102 for V2>
  - Select “Save”
- If all done correctly, the connection status of the outgoing connection should say “CONNECTED” and the group should be green within the Federation Hub

## Edit Outgoing Connection

|                             |                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| *Display Name               | <input type="text" value="FEDHUB"/>                                                                                                              |
|                             | <small>Display name may contain upper and lower case letters, numbers, spaces and underscores up to 30 characters.</small>                       |
| *Address                    | <input type="text" value="192.168.1.126"/>                                                                                                       |
|                             | <small>Provide a valid hostname or IPv4 address.</small>                                                                                         |
| *Port                       | <input type="text" value="9102"/>                                                                                                                |
|                             | <small>Port must be an integer between 1 and 65535.</small>                                                                                      |
| *Reconnect Interval         | <input type="text" value="30"/>                                                                                                                  |
|                             | <small>The number of seconds between connection attempts. A value of 0 will disable automatic reconnect.</small>                                 |
| Max Retries Unlimited       | <input checked="" type="checkbox"/>                                                                                                              |
| Max Retries                 | <input type="text" value="0"/>                                                                                                                   |
|                             | <small>How many times to retry reconnecting before fallback or stopping. Checking</small>                                                        |
| Enabled                     | <input type="text" value="True"/>                                                                                                                |
| Protocol Version            | <input type="text" value="2 - 1.3.11 and Later (Federated Data Sync Sup"/>                                                                       |
| Fallback Connection         | <input type="text" value="No Fallback"/>                                                                                                         |
| Federation Connection Token | <input type="text"/>                                                                                                                             |
|                             | <small>This field is used for creating a federation connection using token authentication rather than X509 client mutual authentication.</small> |

\*Required

# TAK Server - Federation Hub - Connection Walkthrough

## Active Connections

| Federate Configuration | Remote Address | Port | Initiator | Read Count | Processed Count | Groups | Group Mappings (Remote:Local) | Manage                                                                                                                                                 |
|------------------------|----------------|------|-----------|------------|-----------------|--------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| takserver:TAK-ID-CA-01 | 192.168.1.123  | 9102 | Self      | 1          | 0               |        |                               | <a href="#">View Contacts</a><br><a href="#">Delete</a><br><a href="#">Edit Config</a><br><a href="#">Edit Groups</a><br><a href="#">Edit Missions</a> |

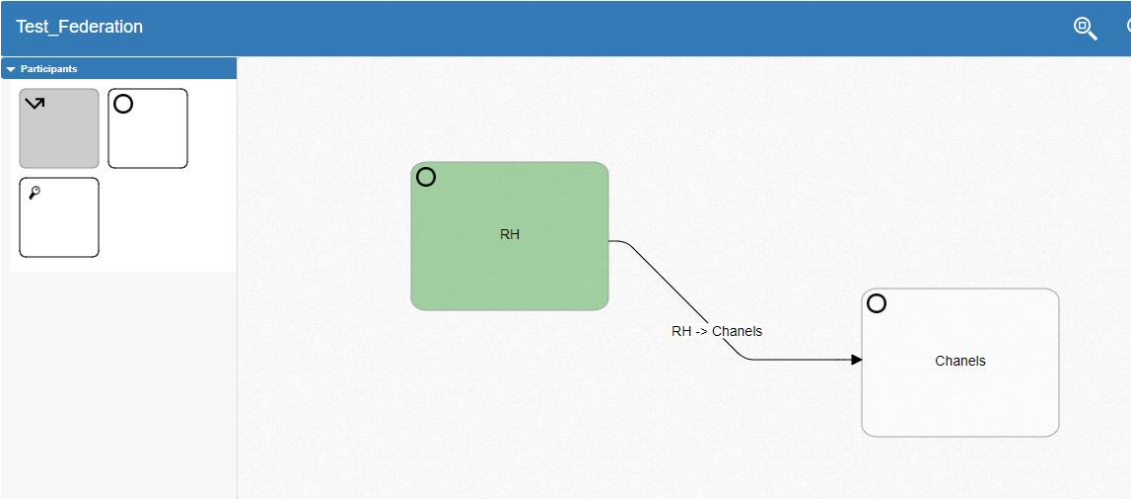
## Outgoing Connection Configuration

[Create Outgoing Connection](#)

| Name | Address       | Port | Fallback Connection | Status                                          | Reconnect Interval | Max Retries | Federate               | Protocol Version | Connection Status | Last Error |                      |                        |
|------|---------------|------|---------------------|-------------------------------------------------|--------------------|-------------|------------------------|------------------|-------------------|------------|----------------------|------------------------|
| test | 192.168.1.123 | 9102 |                     | Enabled<br><a href="#">(Disable Connection)</a> | 30                 | Unlimited   | takserver:TAK-ID-CA-01 | 2                | CONNECTED         |            | <a href="#">Edit</a> | <a href="#">Delete</a> |

## Federate Configuration

| Name                   | Share Alerts | Archive | Notes |                                                                                                                       |
|------------------------|--------------|---------|-------|-----------------------------------------------------------------------------------------------------------------------|
| takserver:TAK-ID-CA-01 | Enabled      | Enabled |       | <a href="#">Delete</a><br><a href="#">Edit Config</a><br><a href="#">Edit Groups</a><br><a href="#">Edit Missions</a> |



# TAK Server - Federation Hub - Outgoing Connections

- An Outgoing connection can be used to connect the federation hub to other hubs or TAK Servers.
- To create an Outgoing connection, drag and drop the outgoing arrow onto the graph.
- Once placed, you will be prompted to give it a name, host, port and enabled option.
- There are three states for the connection status:
  - **Grey:** Disabled
  - **Red:** enabled but not connected
  - **Green:** enabled and connected
- To direct the flow, you can select the outgoing connection and use the arrow to connect nodes.



# TAK Server – Updating Federation Hub

- Before updating the hub, it is recommended that you back up the policy file and list of authorized users.
  - **mv /opt/tak/federation-hub/ui\_generated\_policy.json /tmp**
  - **mv /opt/tak/federation-hub/authorized\_users.yml /tmp**
  - You can replace the files after the upgrade by running the following:
    - **mv /tmp/ui\_generated\_policy.json /opt/tak/federation-hub/**
    - **mv /tmp/authorized\_users.yml /opt/tak/federation-hub/**
- Update Federation Hub:
  - Download and copy over an update version of Federation Hub from your local machine to the server.
  - Run the following to update:
    - **sudo yum upgrade takserver-fed-hub-\*.noarch.rpm**